

University of Pittsburgh

Introduction to Operating Systems
CS 1550

Spring 2022

Sherif Khattab

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(Some slides are from **Silberschatz, Galvin and Gagne ©2013)**

Course Goal and Objectives

Continue to demystify a good portion of the **magic** about how computers work so that you can write more **efficient** programs

The specific objectives are:

1. Modify and compile the Linux kernel to **add system calls**
2. Write **multi-process/multi-thread** programs free from race conditions and deadlocks
3. Simulate page-replacement algorithms for **virtual memory** management
4. Implement a user-land **file system**

Contact Info

- **Course website:** http://www.cs.pitt.edu/~skhattab/cs1550/
- **Instructor:** Sherif Khattab ksm73@pitt.edu

OH: https://khattab.youcanbook.me

TuTh: 16:30-19:30 (remote until Jan 26 then in-person/remote)

F: 16:00-18:00 (**remote only for the entire term**)

- 6307 Sennott Square, Virtual Office: https://pitt.zoom.us/my/khattab

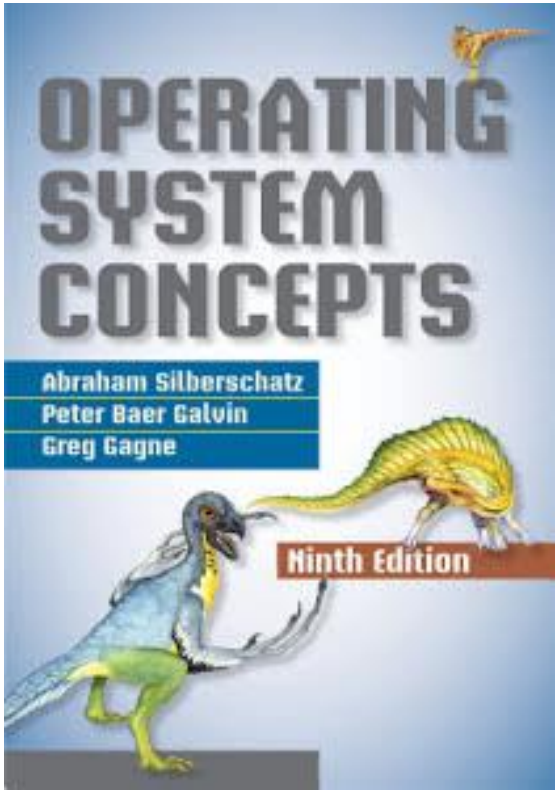
- **Teaching Team:**

TBD

- Recitations start this week!
- **Communication**

1. End of week newsletter
2. Piazza (**Please expect a response within 72 hours)**
    1. Email not recommended!

Textbook



Operating System Concepts (9th Edition)

Silberschatz, Galvin, and Gagne

Grading

- **40%** on 4 projects
  - System programming using C
  - Posted on **Canvas** and submitted on **Gradescope (auto-graded)**
- **20%** on two exams: 12% on higher and 8% on lower
- **18%** on weekly homework assignments
- **14%** recitation: 5 labs using MIT's Xv6 operating system and 4 quizzes on the projects
- **8%** in-class Tophat questions and discussions

Canvas Walkthrough

- Lectures posted on Tophat by end of each lecture day
- Lecture and recitation recordings under Panopto Video
- Pre- and Post-course test
- Piazza for discussion
- Gradescope and autograding policies
- Homework attempts
- Academic Integrity
- NameCoach

Expectations

- Your continuous feedback is important!
  - Anonymous Qualtrics survey
  - Midterm and Final OMET
- Your engagement is valued and encouraged.

Lecture structure (mostly)

Time	Description
~5 min before and after class	Informal chat
~25 min	Announcements, review of reflections, and QA on projects/labs/homework problems
~40 min	Lecturing
~5 min	Tophat questions and/or activities
~5 minutes	QA and muddiest points/reflections

Why is this class (notoriously) hard?

- Lots of concepts
  - Attend lectures and recitations (if you absolutely cannot attend live, watch the video recordings)
  - Study often!
  - Put effort into the weekly homework assignments
- Projects are relatively hard
  - Refresh your C programming and GDB debugging skills (CS 0449!)
  - Start early and show up to office hours!